1 Overview

TP-200 Booster Pump Skid - Operation and Maintenance Manual (fictional test machine)

The TP-200 skid raises municipal water pressure for a small building. It has exactly two main components:

- the inlet cartridge filter (F1), which removes particles before the pump;
- the booster pump (P1), a single-stage centrifugal pump driven by a 1.5 kW motor.

Flow path: supply inlet -> inlet filter F1 -> booster pump P1 -> discharge outlet.

1.1 Rated operating conditions

Rated flow: 6.0 m3/h (permitted range 2.0 to 8.0 m3/h)
Inlet pressure: 1.5 bar (permitted range 0.8 to 3.0 bar)
Discharge pressure: 4.5 bar (permitted range 3.5 to 6.0 bar)
Water temperature: 5 to 35 degC
Duty: continuous, up to 24 operating hours per day

1.2 Monitoring

The controller reports the following channels once every 10 seconds:

inlet_pressure (bar), range 0 to 10
discharge_pressure (bar), range 0 to 10
filter_dp (bar), differential pressure across F1, range 0 to 2
flow_rate (m3/h), range 0 to 12
motor_current (A), range 0 to 10
motor_temperature (degC), range 0 to 120
vibration_rms (mm/s), measured on the pump bearing housing, range 0 to 20

2 Inlet cartridge filter F1

2.1 Function

F1 is a 20 micron pleated cartridge. Particles collect on the cartridge during operation, so the differential pressure filter_dp rises with the volume filtered.

2.2 Limits

Clean cartridge filter_dp at rated flow: 0.10 bar
Warning W-21 "Filter clogging": filter_dp above 0.60 bar
Alarm A-22 "Filter blocked": filter_dp above 0.80 bar. Replace the cartridge.

2.3 Service

Replace the cartridge every 2000 operating hours, or earlier when alarm A-22 is active. Typical cartridge life at rated flow and normal water quality is 2000 to 3000 operating hours.
High particle load (for example after mains repair work) can shorten life to 500 operating hours.

2.4 Effect on the pump

A clogged filter lowers the pump inlet pressure. Below 0.5 bar inlet pressure the pump may cavitate, which raises vibration_rms and accelerates impeller wear.

3 Booster pump P1

3.1 Function

P1 is a single-stage centrifugal pump. Its impeller and wear ring wear slowly in service, which lowers discharge pressure and raises motor current at a given flow. The pump bearings are sealed for life.

3.2 Limits

New pump discharge pressure at rated flow: 4.5 bar

Warning W-31 "Pump performance low": discharge pressure below 3.8 bar at rated flow

Alarm A-32 "Pump worn": discharge pressure below 3.5 bar at rated flow

Motor current at rated flow: 3.2 A nominal, alarm A-33 above 4.5 A

Motor temperature: alarm A-34 above 95 degC

Vibration: warning W-35 above 4.5 mm/s, alarm A-36 above 7.1 mm/s

3.3 Service

Inspect impeller and wear ring every 8000 operating hours.

Expected impeller life at rated conditions: 20000 operating hours.

Bearing rated life (L10): 30000 operating hours.

Operating with cavitation (inlet pressure below 0.5 bar) can halve impeller life.

3.4 Troubleshooting

A-32 with normal filter_dp: worn impeller, schedule pump overhaul.

A-32 with high filter_dp: replace the F1 cartridge first, then re-check.